Advanced Integrated Wet-Lab and Computational Researcher, Levi Lab

UT Southwestern Medical Center · Dallas, TX, US

Job ID (this archive)	HI-027
Employer	UT Southwestern Medical Center
Location(s)	Dallas, TX, US
Career stage (archive tag)	Research / graduate degree
Location group	DFW area
Majors (archive tags)	Biomedical Engineering, Computer Science, Data Science
Pay listed	Not listed
Employment type	Full-time
Employer requisition	968584
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://jobs.utsouthwestern.edu/job/23767005/advanced-integrated-wet-lab-and-computational-researcher-levi-lab-dallas-tx/
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Course sheet refs	HI-027

Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

Computational Biologist II, Advanced Integrated Wet-Lab and Computational Researcher, Levi Lab

WHY UT SOUTHWESTERN?

With over 75 years of excellence in Dallas-Fort Worth, Texas, UT Southwestern is committed to excellence, innovation, teamwork, and compassion. As a world-renowned medical and research center, we strive to provide the best possible care, resources, and benefits for our valued employees. Ranked as the number 1 hospital in Dallas-Fort Worth according to U.S. News & World Report , we invest in you with opportunities for career growth and development to align with your future goals. Our highly competitive benefits package offers healthcare, PTO and paid holidays, on-site childcare, wage, merit increases and so much more. We invite you to be a part of the UT Southwestern team where you'll discover a culture of teamwork, professionalism, and a rewarding career!

JOB SUMMARY

Perform professional work in support of scientific research using technical knowledge of software, software development, networking, and/or hardware in a complex computing environment. Subject matter expert in multiple areas of computing technology, and specific knowledge of molecular biology and genetics. Work collaboratively with Principal Investigator ("PI") to determine the most suitable computational methods and tools to analyze large-scale biological data sets. Prepare reports for presentation or publication. Responsible for assisting PI in meeting the system needs for multiple research projects and/or labs.

This position is designed for an advanced scientist who can bridge wet-lab experimentation and independent computational analysis. The Computational Biologist II will help lead mechanistic and translational studies of injury, fibrosis, regeneration, aberrant mesenchymal cell fate, wound healing, tumor biology, and therapeutic response. The individual will work from study design and sample generation through data preprocessing, integration, biological interpretation, validation, and publication.

LEVI LABORATORY RESEARCH PROGRAM

The Levi Laboratory studies how immune, stromal, vascular, lymphatic, neural, epithelial, and mesenchymal cell populations interact after injury and in disease. The laboratory integrates transgenic and surgical animal models, human biospecimens,

molecular and cellular biology, quantitative imaging, and multiomic technologies to identify mechanisms and therapies that promote faithful tissue repair.

Traumatic heterotopic ossification and aberrant mesenchymal differentiation after burns, musculoskeletal trauma, surgery, and tendon injury.

Fibrosis, wound healing, burn and scar biology, including inflammatory and epigenetic regulation of keratinocyte, fibroblast, immune, and progenitor-cell function.

Tendon, muscle, bone, and soft-tissue repair, including mechanotransduction, extracellular matrix remodeling, and stem or progenitor-cell fate.

Neurovascular and neuroimmune regulation of regeneration, tendinopathy, heterotopic ossification, pain, and osteosarcoma, and adrenergic signaling.

Lymphatic regulation of inflammation, tissue repair, and bone resorption, including VEGF-C or VEGF-D signaling and lymphatic endothelial-cell biology.

Immunometabolism and obesity-related injury responses, including lipid metabolism, inflammatory signaling, and altered regenerative cell states.

Circulating mesenchymal progenitor cells and liquid-biopsy approaches for early diagnosis, risk prediction, and treatment monitoring.

Cancer biology and osteosarcoma, including tumor heterogeneity, tumor microenvironment interactions, angiogenesis, pain, drug response, and pulmonary metastasis.

ESSENTIAL JOB DUTIES

Design integrated experimental and computational strategies with the PI, clinical collaborators, and laboratory scientists, including sample selection, controls, replication, power considerations, batch mitigation, and validation plans.

Independently analyze scRNA-seq, snRNA-seq, bulk RNA-seq, snATAC-seq or other epigenomic data, spatial transcriptomic data, proteomic and metabolomic data, CyTOF or high-parameter flow data, and quantitative imaging datasets from raw files through biological interpretation.

Perform quality control, normalization, batch correction, cell-type annotation, differential analysis, pathway enrichment, trajectory and pseudotime analysis, gene-regulatory network inference, cell-cell communication modeling, ligand-receptor analysis, and cross-modality integration.

Develop spatial and image-computational workflows for immunofluorescence, histology, multiplex imaging, micro-CT, confocal or whole-slide microscopy, and three-dimensional tissue datasets.

Integrate experimental data with clinical variables, longitudinal outcomes, public datasets, and large multiomic databases to identify biomarkers, mechanisms, and therapeutic targets.

Develop and rigorously evaluate statistical and machine-learning models for disease classification, early detection, risk prediction, treatment response, and biomarker prioritization, with appropriate feature selection, cross-validation, calibration, and external or biological validation.

Build, document, test, and maintain reproducible analysis pipelines in R and Python using version control, workflow management, containerized environments, and high-performance or cloud computing resources.

Establish laboratory standards for metadata, data dictionaries, sample tracking, file organization, quality metrics, code review, backup, controlled access, and FAIR data practices while complying with institutional and sponsor requirements.

Perform or directly support wet-lab studies required to generate and validate computational findings, including animal handling, tissue harvest, cell and tissue dissociation, histology, immunostaining, imaging, cell culture, flow cytometry, nucleic acid isolation, PCR or qPCR, and sequencing-library workflows.

Support transgenic colony studies, lineage-tracing experiments, injury and disease models, therapeutic perturbations, and collection of blood, skin, tendon, muscle, bone, tumor, lymphatic, and metastatic tissues.

Translate computational hypotheses into targeted validation experiments using molecular assays, immunohistochemistry, CRISPR-Cas9 or other gene perturbation approaches, pharmacologic studies, and functional cellular or animal assays in collaboration with experimental scientists.

Generate publication-quality visualizations and clear data narratives; lead computational sections of manuscripts, abstracts, grants, progress reports, and presentations; and contribute to high-impact peer-reviewed publications.

Present findings at laboratory meetings and national or international scientific forums, communicate limitations and uncertainty clearly, and respond constructively to scientific review.

Mentor trainees and staff in computational biology, experimental design, statistics, reproducible research, and biological interpretation, while contributing to a respectful, inclusive, and accountable laboratory culture.

STRONGLY PREFERRED TECHNICAL EXPERTISE

Advanced proficiency in R and Python for biological data analysis, statistics, visualization, and pipeline development; experience with Linux, shell scripting, Git, and high-performance computing.

Demonstrated hands-on experience analyzing single-cell transcriptomic datasets independently from raw-data processing and quality control through cell annotation, mechanistic interpretation, figure generation, and validation.

Experience with spatial transcriptomics and at least one additional high-dimensional modality such as metabolomics, proteomics, epigenomics, CyTOF, imaging mass cytometry, or multiplex tissue imaging.

Experience managing and integrating large clinical, imaging, or multiomic databases with robust metadata, reproducible provenance, and appropriate privacy and access controls.

Strong foundation in molecular and cell biology, genetics, cancer or musculoskeletal biology, immunology, developmental biology, regenerative medicine, or a related field.

Practical wet-lab experience with mammalian cell culture, animal handling, tissue processing, embedding, sectioning and staining, immunofluorescence or immunohistochemistry, microscopy, molecular assays, and sample preparation for sequencing or other omic platforms.

Ability to choose appropriate statistical methods, recognize confounding and batch effects, distinguish exploratory from confirmatory analyses, and communicate uncertainty without overinterpreting results.

PREFERRED DOMAIN AND METHODOLOGICAL EXPERIENCE

Mesenchymal progenitor and stem-cell biology, lineage tracing, aberrant cell fate, osteogenesis, chondrogenesis, fibrosis, wound repair, or extracellular matrix biology.

Immune-stromal, neurovascular, lymphatic, or tumor microenvironment analysis, including macrophage states, endothelial populations, sensory and sympathetic nerves, and cell-cell signaling.

Mouse models of burn and musculoskeletal injury, heterotopic ossification, tendon degeneration, obesity or metabolic disease, wound healing, cancer, or metastasis.

Human translational studies using biospecimens, circulating rare-cell populations, biomarker discovery, diagnostic modeling, or longitudinal clinical outcomes.

Tools such as Seurat, Scanpy, Bioconductor, Signac, ArchR, Cell Ranger, Space Ranger, Harmony, scVI, Monocle, Slingshot, CellChat, NicheNet, SCENIC, DESeq2, edgeR, limma, or equivalent platforms.

Quantitative imaging and pathology tools such as QuPath, ImageJ or Fiji, CellProfiler, HALO, Imaris, or custom computer-vision workflows.

SCIENTIFIC LEADERSHIP AND PROFESSIONAL COMPETENCIES

Ability to manage several projects, set priorities, document decisions, meet deadlines, and move analyses from exploratory work to publication-quality completion.

Excellent critical thinking, troubleshooting, scientific writing, oral communication, collaboration, and project-management skills.

Ability to explain computational concepts to experimental and clinical collaborators and translate biological questions into rigorous analytic plans.

Commitment to research integrity, reproducibility, responsible authorship, data stewardship, laboratory safety, continuous learning, and sustained engagement with both bench and computational work rather than functioning exclusively as a data-analysis service.

BENEFITS

UT Southwestern is proud to offer a competitive and comprehensive benefits package to eligible employees. Our benefits are designed to support your overall wellbeing, and include:

PPO medical plan, available day one at no cost for full-time employee-only coverage

100% coverage for preventive healthcare-no copay

Paid Time Off, available day one

Retirement Programs through the Teacher Retirement System of Texas (TRS)

Paid Parental Leave Benefit

Wellness programs

Tuition Reimbursement

Public Service Loan Forgiveness (PSLF) Qualified Employer

Learn more about these and other UTSW employee benefits!

EXPERIENCE AND EDUCATION

Required

Education

PhD in Computer Science or a related field of biological science, with thesis work in bioinformatics and computational biology or

Master's Degree in Computer Science or a related field of biological science or

Bachelor's Degree in Computer Science or a related field of biological science

Experience

2 years of related research experience in bioinformatics and computational biology with Master's Degree or

4 years of related research experience in bioinformatics and computational biology with Bachelor's Degree.

JOB DUTIES

Manage computer hardware and software programming and maintenance for one or more research laboratories. Ensure existing systems meet the continuing needs of the labs and/or recommend and test new systems as necessary.

Perform or direct others to perform complex data analysis related to specialized research methodologies and results. Manage assigned research projects with minimal input from PI.

Develop and/or modify software to support new and ongoing research projects.

Resolve hardware, software, and network issues in support of the lab.

Oversee the development and maintenance of complex databases to support new and ongoing research projects.

Prepare research reports for presentation, and assist with preparation of manuscripts for publication.

May specialize in bioinformatics, software development, network administration, or hardware/software maintenance.

Perform other duties as assigned.

SECURITY AND EEO STATEMENT

Security

This position is security-sensitive and subject to Texas Education Code 51.215, which authorizes UT Southwestern to obtain criminal history record information.

EEO

UT Southwestern Medical Center is committed to an educational and working environment that provides equal opportunity to all members of the University community. As an equal opportunity employer, UT Southwestern prohibits unlawful discrimination, including discrimination on the basis of race, color, religion, national origin, sex, sexual orientation, gender identity, gender expression, age, disability, genetic information, citizenship status, or veteran status.

Notes for ECS 1100 students

How this connects to your major

Biomedical Engineering: device design, clinical use, testing, quality, human factors or biomedical research; Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology. These are teaching annotations, not an employer guarantee of eligibility.

A path to explore after further study; graduate-level requirements are retained.

BME: Teaching interpretation of role tasks; major eligibility is not explicitly confirmed

DS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, R, Git/source control, Statistics, Data visualization, Cloud, Verification/validation, Genomics/bioinformatics, Communication/teamwork

Eligibility excerpts not shown above

Compensation (source excerpts): Not stated in captured posting text.

Course sheet tags

Majors (teaching tags): BME; DS; Career stage (teaching tag): Research / graduate training path; Geography: DFW; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full-time.

Source record

Original posting https://jobs.utsouthwestern.edu/job/23767005/advanced-integrated-wet-lab-and-computational-researcher-levi-lab-dallas-tx/
Accessed UTC: 2026-10-07T20:11:33.841593+00:00
Saved raw source: sources/postings/utsw_23767005.json
SHA-256: ce50985af35a9527baf808c021554817b9a969a38f3c04aef4ceadee7265317f